INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

(School of University of Engineering and Management, Kolkata)

End Semester Examination, April – May, 2025

Degree: **BCA** Semester : **Second**

Stream: **BCA**

Paper Code: **BCAMD201**

Paper Name: **Mathematics for Computing**

Full Marks: **100** Duration: **3 Hours**

Part - A

Attempt any 10 questions out of 15 questions

Each question carries 2 Marks (2 × 10)

1. If A and B be events with $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{4}$, and $P(A \cup B) = \frac{1}{2}$ then find $P(B/A)$? **2**

2. Find the A.M of the variable assuming the data 8,5,3,10,12. **2**

3. Calculate the H.M of 1, $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$. **2**

4. In a single cast with two dice the chance that one die turns up 3 and the other 4 is ? **2**

5. The probability that Pradyut passes is 0.9 and Subrata passes is 0.8. The probability that at least one of them passes is ? **2**

6. Find the First and Third Quartile of the following datas : **2**

   2, 5, 8, 9, 12, 17, 19, 21, 27, 31, 33

7. Find the A.M. of the following frequency distribution : **2**

x	:	6	5	2	8
Frequency	:	4	3	1	2

8. The A.M. of two observations is 25 and their HM is 9; find their G.M. **2**

9. The mean of the following distribution is ? **2**

Marks	: 20 – 39	40 – 59	60 – 79	80 – 99
No. of Students	: 10	12	8	10

10. The median of the following distribution is ? **2**
10, 13, 9, 7, 37, 16, 27, 32

11. If the relations between two set of observations $\{x_i\}$ and $\{y_i\}$ is $2y_i - 6x_i = 6$ and mode of the first set is 21 then the mode of the second set is ? **2**

12. The variance of the following distribution is ? **2**

x_i	:	-1	0	3	4
f_i	:	3	2	1	4

13. $2x_i + y_i = 3$ is the relation between two sets of datas $\{x_i\}$ and $\{y_i\}$. If $\sigma_x = 3$ then $\sigma_y = ?$ **2**

14. For $n = 0$, $\sum x = 60$, $\sum x^2 = 1000$ the s.d = ? **2**

15. Quartile deviation ignores 50% of the data. TRUE or FAlSE? Explain. **2**

Part - B

Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

16. Two urns contain respectively 2 red, 5 black, 7 green and 1 red, 4 black, 9 green balls. One ball is drawn from each urn. Find the probability that both the balls are of the same colour. **5**

17. The face cards are removed from a pack of 52 cards. Then 4 cards are drawn one by one from the remaining 40 cards. What is the probability that 4 cards belong to different suits and different denomination? **5**

18. Represent the following data by vertical bar diagram: **5**

Year	:	1976-77	1977-78	1978-79	1979-80	1980-81	1981-82
Income	:	20	40	30	60	50	80

19. Represent the following data by a Line Chart choosing suitable scales.The daily profit and losses of a business concern for the first 10 days of a month are given in the following table : **5**

Day of the month:	1	2	3	4	5	6	7	8	9	10
Profit in Rs. :	400	800	600	900	200	200			800	1000
Loss in Rs. :							400	100		

20. Draw Histogram representing the following data : 5

Marks obtained in Mathematics by candidates is given by the following frequency distribution:

Marks	:20-25	25-35	35-45	45-60	60-70	70-75	75-80
No. of candidates :	4	16	18	15	12	5	6

21. Find D_7 from the following frequency distribution of marks: 5

Marks	: 23	25	26	30	31	33	34	35	37	40
No. of students	: 10	20	5	15	25	12	13	14	9	6

22. The mean monthly salary paid to all employees in a certain company was Rs. 600. The mean monthly salaries paid to the male and female employees was Rs. 620 and Rs. 520 respectively. Find the percentage of male and female employees in the company. 5

23. Construct the **Frequency Polygon** for the following frequency distribution: 5

Weights: (in kg)	41-45	46-50	51-55	56-60	61-65	66-70	71-75	76-80
No. of men :	4	5	9	6	11	5	7	3

24. Let x be a variable which assumes the datas: 20,85,120,60,40. Then $\bar{x} = \frac{1}{5}(20 + 85 + 120 + 60 + 40) = 65$. Find variance and S.D. 5

Part - C

Attempt any 5 questions out of 8 questions

Each question carries 10 Marks (10 × 5)

25. In an examination 30% of the students failed in Physics, 25% in Mathematics and 12% in both Physics and Mathematics. A student is selected at random. Find the probability that 10

(i) the student has failed in Physics, if it is known that he has failed in Mathematics.

(ii) the student has failed at least one of the two subjects

(iii) the student has passed at least one of the two subjects.

(iv) the student has passed in Mathematics if he failed in Physics.

26. 15 new students are to be evenly distributed among 3 classes. Suppose that there are 3 whiz-kids among the fifteen. What is the probability that each class gets one whiz-kid and one class gets them all? 10

27. Draw the **Ogive** (less than and more than type) for the following distribution: **10**

Mid point	18	25	32	39	46	53	60	Total
Frequency	10	15	32	42	26	12	9	146

28. The A.M. calculated from the following frequency distribution is known to be 72.5. Find the value of x : **10**

Classes	30-39	40-49	50-59	60-69	70-79	80-89	90-99
Frequency	2	3	11	20	x	25	7

29. The median and mode of the following frequency distribution are known to be 27 and 26 respectively. Find the values of α and β : **10**

Class interval	0-10	10-20	20-30	30-40	40-50
Frequency	3	α	20	12	β

30. Find the variance and standard deviation of the following frequency distribution : **10**

Weight (in kg)	36-40	41-45	46-50	51-55	56-60	61-65	66-70
No of persons	14	26	40	33	50	37	25

31. The mean and the standard deviation of marks of 70 students were found to be 65 and 5.2 respectively. Later it was detected that the value 85 was recorded wrongly and therefore it was removed from the data set. Then find the mean and s.d for the remaining 69 students. **10**

32. Distribution of wages in two factories A and B : **10**

Wages in Rs	50-100	100-150	150-200	200-250	250-300	300-350
No. of workers in A	2	9	29	54	11	5
No. of workers in B	6	11	18	32	27	11

State in which factory the wages are more variable by evaluating respective C.V.

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